PRODUCTION BLUEPRINT & EXTENSION GUIDE

Study Buddy: Royal 3D

Complete Architectural Summary, UI Breakdown, Data Engine & LLM Extension Guide

1. Project Identity & Core Specs

Property	Value & Specification
Project Name	`study-buddy-royal`
Core Stack	Next.js 15 (App Router), Tailwind CSS v4, Framer Motion, Zustand v5, Firebase v10
Design System	Royal Obsidian (`#090D16`) + Frosted Glass (blur 20px, saturate 180%) + 3D Tactile Buttons
Viewport	Mobile-first (max-width 480px centered), 100dvh zero-scroll quiz theater
Target Audience	NCERT Class 10 (Delhi Hinglish Edition) across 5 subjects × 13 chapters

2. 5-Subject Theme Colors & Dynamic Glow

Subject	ID	Theme Color	3D Accent	Chapters
Science (विज्ञान)	`sci`	#10B981 (Emerald)	#047857	13 Chapters (Ch 1–13)
Math (गणित)	`math`	#3B82F6 (Royal Blue)	#1D4ED8	13 Chapters (Ch 1–13)
SST (सामाजिक विज्ञान)	`sst`	#F59E0B (Amber Gold)	#B45309	13 Chapters (Ch 1–13)
Hindi (हिंदी)	`hin`	#F43F5E (Rose)	#BE123C	13 Chapters (Ch 1–13)
English (अंग्रेज़ी)	`eng`	#8B5CF6 (Violet)	#6D28D9	13 Chapters (Ch 1–13)

3. Complete UI Breakdown (Screens & Navigation)

A. 4-Tab De-Bloated Bottom Dock

- **Path (Home):** S-curve winding path, sticky chapter banners, 72px tactile nodes with lock/unlock states.
- **Mistakes (Smart Mistake Vault):** Filter pills (All, Sci, Math, SST, Hindi, Eng), re-attempt questions, instant removal upon correct solve.
- **Leaderboard:** League divisions (Diamond, Gold, Silver, Bronze), podium spotlight for Top 3, highlighted user row.
- **Profile:** Royal ATM/credit-card hero card, 2×2 Bento performance radar (Streak, XP, Accuracy, Crowns), hearts refill.

B. Distraction-Free Quiz Theater (`100dvh`)

- **Top HUD:** Close button `[X]`, liquid progress spring bar with tip sparkle, Sound mute toggle, Hearts counter (5 max, 3 on Boss).
- **Circular Timer:** Countdown ring (30s on L2, 20s on L3, 15s on L4, 12s on L5 Boss). Heartbeat pulse at ≤5s.
- **Question Card:** 2-line max, 18–20px bold font, Devanagari line-height 1.65, visual difficulty dots (`●●●●○ Level 4`).
- **Tactile Option Bars:** 4 full-width 3D bars with `[A][B][C][D]` 3D keycaps. Keyboard shortcuts: `1, 2, 3, 4` to select, `Enter` to check.

- **50:50 Lifeline:** Blurs and disables 2 wrong options (`opacity: 0.25, scale: 0.95`).
- **Contextual Explanation Drawer:** Spring-elastic bottom sheet (`stiffness: 380, damping: 28`) with 1-line reason + `CONTINUE →` .
- **Result & Victory Screen:** 3-Star sequential impact stamp ("THUD"), XP slot-machine odometer roll, 3D Gold Crown on Boss clear, shareable WhatsApp postcard.

4. Zero-Credit Static Data Architecture

> **Zero Token Burn Strategy:** Questions are stored as static JSON files in `/public/data/` and cached in client `localStorage`. Zero AI API calls at runtime!

Folder Hierarchy

```
/public/data/
├── manifest.json              # 5 subjects × 13 chapters index metadata
├── /science/ch_01.json ... ch_13.json
├── /math/ch_01.json ... ch_13.json
├── /social_science/ch_01.json ... ch_13.json
├── /hindi/ch_01.json ... ch_13.json
└── /english/ch_01.json ... ch_13.json
```

Ultra-Compact Question Schema (Single-Letter Keys)

```
{
  "i": 101,                          // Unique Question ID
  "d": 1,                            // Difficulty: 1=Easy, 2=Moderate, 3=Speed, 4=Hard, 5=Boss
  "q": "प्रकाश वर्ष किसका मात्रक है?",     // Question statement
  "o": ["समय", "दूरी", "चाल", "बल"],     // Array of exactly 4 options
  "a": 1,                            // 0-indexed correct option (0=A, 1=B, 2=C, 3=D)
  "e": "प्रकाश द्वारा 1 वर्ष में तय की गई दूरी।" // 1-line explanation (max 12 words)
}
```

5. Gamification & Strict Lock State Machine

Level	Node Type	Timer	Hearts	Unlock Rule
Level 1	Bronze Circle	No Timer	5	Default Unlocked (Ch 1). 10 basic questions.
Level 2	Silver Circle	30s/Q	5	Unlocks ONLY if Level 1 score ≥ 75%. 12 questions.
Level 3	Gold Circle	20s/Q	5	Unlocks if Level 2 ≥ 75%. 1.5×/2×/3× Streak Multiplier active.
Level 4	Platinum Circle	15s/Q	5	Unlocks if Level 3 ≥ 75%. Numerical & tricky statements.
Level 5	Diamond Hex Shield	12s/Q	3 ONLY!	Boss Level. Clears Chapter, awards 3D Crown, unlocks Chapter M+1.

6. State Management & Firebase Schema

Zustand Client Stores

- `useUserStore` : User profile, XP, streak, hearts, active subject, mistake resolution.

Single-Doc Firestore (`/users/{uid}`)

```
{
  "name": "Aarav Sharma",
  "email": "aarav@gmail.com",
  "xp": 1450,
```

- `useQuizStore` : Current question, selected option, 50:50 lifeline, timer tick, combo streak, result states.
- `useToastStore` : Dynamic Island notifications ("Level Unlocked!").

```
  "streak": 12,
  "lastActive": "2026-11-06",
  "hearts": 5,
  "progress": { "sci_ch01": 5, "sci_ch02": 2 },
  "mistakes": [103, 208, 412]
}
```

* Cloud Sync Strategy: Writes happen **strictly upon level completion** to stay 100% within Firebase Spark Free Tier limits.

7. How to Extend the Application (Guide for LLMs & Devs)

To add more features or content without breaking existing architecture, follow these exact patterns:

A. How to Add New Chapters or Real OMR Questions

1. Convert your OMR questions into the ultra-compact JSON format using prompt:

```
"Convert Chapter [X] questions to minified JSON with keys: i (int), d (1-5), q (str), o ([4 str]), a (0-3 index), e (short explanation)."
```

2. Save the file in `/public/data/{subject}/ch_{numPad}.json` (e.g. `ch_03.json`).
3. Register the chapter in `/public/data/manifest.json` under the respective subject.

B. How to Add New Game Power-Ups

- Open `store/useQuizStore.ts`: Add the power-up state (e.g., `streakFreeze`, `hintToken`, `timeFreeze`).
- Add a button in `components/quiz/QuizTheater.tsx` next to the `50:50` button.
- Deduct XP or coins in `useUserStore` upon activation.

C. How to Add New Audio SFX

- Open `lib/audioEngine.ts`: Write a new Web Audio API oscillator function (e.g., for level unlock, badge unlock, or time warning). Zero MP3 downloads required!

D. How to Add a New Tab to the Bottom Dock

- Add the route in `app/(app)/[new_tab]/page.tsx`.
- Add the icon and link in `components/layout/BottomDock.tsx` in the `tabs` array.

8. Verification Gate & Build Status

Verification Step	Command	Result
Static Page Generation	`npm run build`	✓ PASSED — 0 errors across 10 static & dynamic routes
Unit Test Suite	`npm test`	✓ PASSED — 3 test suites, 100% pass rate in 18ms
Visual Design Check	Puppeteer Chrome Headless	✓ PASSED — 4 high-res 3D glassmorphic screenshots verified
Local Server Port	`npx next start -p 3000`	✓ LIVE at http://localhost:3000

Study Buddy: Royal 3D is 100% Production-Ready & Fully Extensible